

DREAMRec: Denoised and Decoupled Retrieval with Adaptive Memory for Sequential Recommendation

Anonymous submission

Abstract

Sequential Recommender Systems aim to capture dynamic user preferences by mining historical dependencies. However, existing methods typically adopt a truncation strategy that focuses mainly on the most recent interactions. From a cognitive psychology perspective, real-world user behavior is jointly driven by immediate intent and accumulated historical habits. The truncation strategy ignores information in the dimension of historical preferences and is limited entirely to recent information. A straightforward solution is to feed the user's entire historical sequence into the model. However, as the sequence length increases, the attention weights for each item are diluted. The dilution confronts two critical challenges: amplified noise interference and the inability to focus on specific regularities. To address these issues, we propose **DHPRec** (**D**enoised **H**istorical **P**atterns for Sequential **R**ecommendation), a novel framework designed to bridge immediate intent and specific long-term patterns. DHPRec introduces a Frequency-Domain Feature Refinement mechanism to filter high-frequency noise and extract stable long-term representations. Furthermore, we design a Pattern-Based Anchor-Guided Fusion mechanism, which extracts pattern units representing long-term regularities and utilizes an anchor to effectively extract the global specific historical preferences.

Introduction

Sequential recommendation aims to predict users' next behaviors based on their historical interactions and has become an essential component in real-world applications, such as e-commerce, short-video platforms, and news recommendation systems. With the growing number of users and the increasing diversity of item content, user behavior sequences have become more complex and heterogeneous. Therefore, how to accurately capture users' interests from complex historical interactions remains a fundamental challenge in sequential recommendation.

Recently, the rapid development of pre-trained language models has driven increasing efforts to leverage textual descriptions of items for semantic representation learning, shifting sequential recommendation from traditional ID-based representations toward semantic-aware modeling. Ex-

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

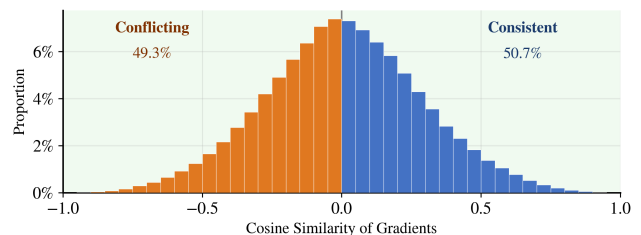


Figure 1: Distribution of Gradient Conflicts between Retrieval and Prediction Tasks.

isting methods typically feed item semantic representations into sequential encoders, such as CNNs, RNNs, and Transformers, to learn a unified user representation from historical interactions, which is then matched with candidate items for ranking. This paradigm of "item semantic encoding-sequential representation learning-user representation matching" has become a prevalent framework in modern sequential recommendation.

However, user historical behaviors inherently contain diverse types of information, including behavioral noise, long-term preferences, and short-term emerging intents. Existing unified user modeling approaches usually compress these heterogeneous signals into a single representation, which may cause interference among different interest patterns and limit the model's ability to accurately characterize users' true preferences. Specifically, existing methods still face the following three challenges.

The first challenge lies in distinguishing behavioral noise from genuine interests. User historical behaviors simultaneously contain various signals, including genuine preferences, exploratory behaviors, and random noise. Existing denoising methods typically assume that high-frequency signals correspond to noise and suppress them accordingly. However, high-frequency components may also reflect users' genuine bursty interests, such as a strong short-term demand for a new category that appears intensively within a short period. Therefore, directly filtering such signals may remove valuable behaviors, making it difficult for models to accurately distinguish noise from true interests.

The second challenge is the lack of dynamic coordination between short-term and long-term interests. Users' next

behaviors are usually driven jointly by long-term stable preferences and short-term immediate needs. Existing methods mainly model different temporal scales implicitly through truncation strategies, attention mechanisms, or hierarchical structures. However, massive repetitive and sparse interactions may obscure core preference patterns, making it difficult to form stable and structured memories. Moreover, when long-term preferences conflict with recent needs, such as a user with a long-term preference for sports suddenly becoming interested in cooking, existing models lack the ability to perceive the current context and dynamically determine which interest should dominate. Therefore, how to dynamically balance different interests while consolidating long-term memory remains an open challenge.

The third challenge is the representation bottleneck caused by unified user representations. Existing methods typically compress user historical behaviors into a single representation, which is then simultaneously used for historical retrieval and recommendation prediction. However, retrieval requires the representation to preserve similarity with historical behaviors, whereas prediction requires the representation to maintain sufficient discriminative capability for distinguishing candidate items. These two objectives are inherently inconsistent. When a single representation is optimized for both tasks simultaneously, conflicts between objectives may arise, limiting the expressive capability of user representations. As shown in Figure 1, on the ML-1M dataset, approximately 49.3% of samples exhibit conflicting gradient directions between the retrieval objective and the prediction objective (cosine similarity < 0), indicating that unified representation learning is indeed affected by interference from different optimization objectives.

To address the above three challenges, we propose DREAMRec (Denoised and Decoupled Retrieval with Adaptive Memory for Recommendation), an adaptive memory modeling framework for sequential recommendation. DREAMRec progressively refines user representations through three stages: behavior denoising, semantic memory construction, and decoupled retrieval. Specifically, we first introduce a Wavelet-Enhanced Burstiness-preserving Denoiser (WEBD), which suppresses behavioral noise while preserving genuine interest signals. Then, we develop a Semantic Memory Prototyping module (SMP) to organize denoised historical behaviors into compact long-term semantic memories, and further incorporate context-aware queries to dynamically coordinate short-term and long-term interests. Finally, we design a Decoupled Memory Retrieval module (DMR), which performs memory matching and user representation learning in separate retrieval and representation spaces, alleviating the objective conflicts caused by unified representation learning. Through these designs, DREAMRec progressively achieves behavior refinement, interest organization, and retrieval optimization, effectively reducing the interference among heterogeneous interest signals and enabling more accurate and robust user modeling.

The main contributions of this paper are summarized as follows:

- We propose DREAMRec, an adaptive memory model-

ing framework for sequential recommendation, which enables more refined user interest modeling through behavior denoising, semantic memory construction, and decoupled retrieval.

- We design a wavelet-enhanced denoising mechanism that preserves bursty interests, along with a memory modeling approach based on semantic prototypes and decoupled retrieval, effectively improving the robustness and discriminability of user representations.
- Extensive experiments on multiple public benchmarks demonstrate that DREAMRec consistently outperforms state-of-the-art methods, and ablation studies further validate the effectiveness of each proposed component.

Method

Problem Formulation

Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{V} = \{v_1, \dots, v_{|\mathcal{V}|}\}$ denote the user set and the item set, respectively. Each item $v \in \mathcal{V}$ is associated with a unique item ID as well as a set of textual attributes, such as the product title, brand, and category. To leverage semantic information, we employ a semantic encoder to obtain the semantic representation of each item, denoted by $z_i \in \mathbb{R}^d$, where d is the embedding dimension.

For each user, the historical interaction sequence is represented as $\mathcal{S} = \{v_1, \dots, v_L\}$, and the corresponding sequence of semantic representations is denoted by $\mathbf{Z} = [z_1, \dots, z_L]$. Given the historical interaction sequence $\mathcal{S}$, the objective of sequential recommendation is to predict the next item that the user is most likely to interact with, i.e., maximizing the conditional probability: $P(v_{L+1} | \mathcal{S})$.

Wavelet-Enhanced Burstiness-preserving Denoiser

High-frequency variations in user behavior sequences may originate from either random noise or genuine bursty interests. Existing frequency-based denoising methods usually suppress high-frequency components uniformly, which may remove informative signals reflecting emerging user preferences. To address this issue, we propose the Wavelet-Enhanced Burstiness-preserving Denoiser (WEBD), which refines behavior representations by suppressing noise while preserving meaningful bursty patterns.

Specifically, WEBD decomposes the behavior sequence into low-frequency trends and high-frequency details via wavelet transform, and learns context-aware thresholds to selectively suppress noisy variations while retaining informative high-frequency signals. The refined sequence is then reconstructed through inverse wavelet transform, providing reliable behavior representations for subsequent semantic memory modeling.

Behavior Decomposition Given the user behavior representation sequence $\mathbf{Z} = [z_1, z_2, \dots, z_L]$, WEBD performs multi-scale decomposition using the Discrete Wavelet Transform (DWT):

$$\mathbf{A}, \mathbf{D} = \text{DWT}(\mathbf{Z}), \quad (1)$$

where $\mathbf{A}$ and $\mathbf{D} \in \mathbb{C}^{(L'+1) \times d}$, $L' = \lfloor L/2 \rfloor$. Here, $\mathbf{A}$ and $\mathbf{D}$ denote the low-frequency trend and high-frequency detail components, respectively. The low-frequency component

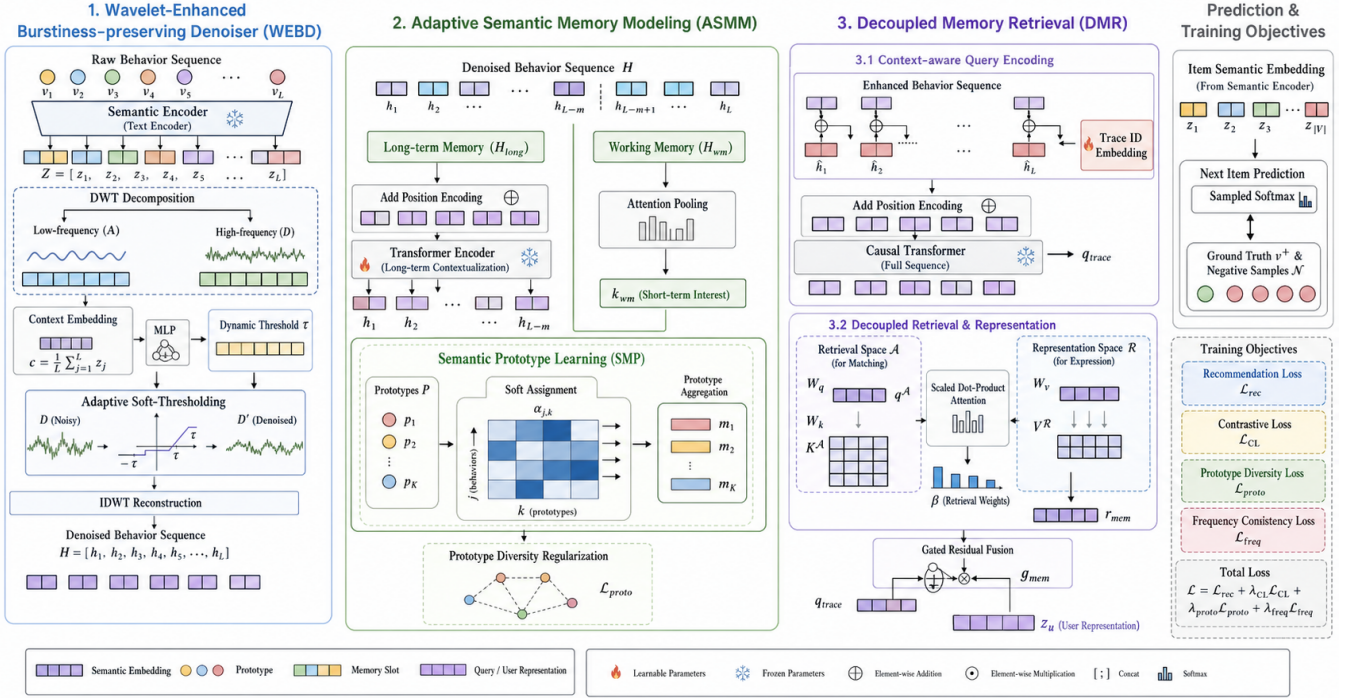


Figure 2: The overall architecture of DHPRec. (Bottom) The Frequency-Domain Feature Refinement module constructs refined item representations via spectral denoising. (Middle) The Pattern-Based Anchor-Guided Fusion Network, which transforms the sequence into compact Pattern Units and utilizes the Anchor to extract relevant long-term regularities. (Top) A motivating example illustrating how the Immediate Intent (e.g., Christmas demand) bridges specific Historical Habits.

captures stable interest evolution, while the high-frequency component reflects local behavioral changes. Since high-frequency variations may represent either noise or genuine bursty interests, adaptive discrimination is required.

Adaptive Burst-preserving Denoising We learn context-dependent thresholds from the entire behavior sequence. Specifically, a global contextual representation is obtained by average pooling:

$$c = \frac{1}{L} \sum_{j=1}^L z_j. \quad (2)$$

The contextual representation is then fed into a lightweight MLP to predict channel-wise adaptive thresholds:

$$\tau = \alpha \cdot \sigma(\text{MLP}(c)), \quad (3)$$

where α is a learnable scaling factor and $\sigma(\cdot)$ denotes the sigmoid function. Given the learned thresholds, soft-thresholding is applied to the high-frequency coefficients:

$$D' = \text{sign}(D) \cdot \max(|D| - \tau, 0). \quad (4)$$

The adaptive threshold mechanism adjusts the denoising strength according to behavioral contexts. Smaller thresholds preserve consistent high-frequency patterns corresponding to bursty interests, while larger thresholds suppress irregular fluctuations caused by noise.

Behavior Reconstruction Finally, the refined high-frequency components are combined with the low-frequency trends and reconstructed through the Inverse Discrete Wavelet Transform (IDWT):

$$H = \text{IDWT}(A, D'). \quad (5)$$

The reconstructed sequence is represented as $H = [h_1, h_2, \dots, h_L]$. By selectively removing random fluctuations while preserving stable trends and bursty interests, WEBD provides cleaner inputs for subsequent semantic memory modeling.

Adaptive Semantic Memory Modeling

Although WEBD refines behavior representations, the resulting sequence still mixes short-term intentions and long-term preferences. To explicitly model interests across different temporal scales, we propose Adaptive Semantic Memory Modeling, which organizes user behaviors into a hierarchical memory structure. Specifically, recent interactions are preserved as Working Memory for capturing immediate interests, while long-term behaviors are compressed into Semantic Prototypes to construct compact and retrievable semantic memories.

Memory Construction Given the refined behavior sequence from WEBD, $H = [h_1, h_2, \dots, h_L]$, To capture

user interests at different temporal scales, we divide the sequence into working memory and long-term memory according to a ratio ρ : $\mathbf{H}_{short} = [\mathbf{h}_{L-\lfloor \rho L \rfloor + 1}, \dots, \mathbf{h}_L]$, $\mathbf{H}_{long} = [\mathbf{h}_1, \dots, \mathbf{h}_{L-\lfloor \rho L \rfloor}]$.

The working memory captures recent user intentions, while the long-term memory models stable preference patterns. For each representation $[\mathbf{H}_{short}]_j$, where $j \in \{1, \dots, |\mathbf{H}_{short}|\}$, we obtain the short-term interest:

$$a_j = \text{softmax}(\mathbf{w}_a^\top [\mathbf{H}_{wm}]_j), \quad (6)$$

$$\mathbf{s} = \sum_{j=1}^{|\mathbf{H}_{short}|} a_j [\mathbf{H}_{short}]_j. \quad (7)$$

where $\mathbf{w}_a$ is a learnable attention vector. The resulting $\mathbf{s}_{wm}$ serves as the short-term interest representation for subsequent memory retrieval.

Semantic Prototype Learning Directly retaining all historical behaviors introduces redundancy and makes it difficult to form reusable long-term memories. Therefore, we compress long-term behaviors into a set of semantic prototypes. We maintain K learnable semantic prototypes $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_K] \in \mathbb{R}^{K \times d}$. For the long-term memory $\mathbf{H}_{long}$, positional embeddings and a Transformer encoder are applied to capture temporal dependencies:

$$\tilde{\mathbf{H}}_{long} = \text{Transformer}(\text{LN}(\mathbf{H}_{long} + \mathbf{P}_{long})). \quad (8)$$

where $\mathbf{P}_{long}$ denotes the learnable positional embeddings. For each contextualized behavior representation $\tilde{\mathbf{h}}_j$, its assignment probability over prototypes is computed as:

$$\alpha_{j,k} = \frac{\exp(\tilde{\mathbf{h}}_j^\top \mathbf{p}_k / \tau_p)}{\sum_{l=1}^K \exp(\tilde{\mathbf{h}}_j^\top \mathbf{p}_l / \tau_p)}. \quad (9)$$

where τ_p is a temperature coefficient controlling the sharpness of prototype assignments. Then the k -th semantic memory slot is obtained by weighted aggregation:

$$\mathbf{m}_k = \frac{\sum_j \alpha_{j,k} \tilde{\mathbf{h}}_j}{\sum_j \alpha_{j,k} + \epsilon}. \quad (10)$$

where ϵ is a small constant for numerical stability. The resulting long-term semantic memory is $\mathbf{M}_{long} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{K \times d}$. By converting numerous historical interactions into a small number of semantic prototypes, the model obtains compact long-term memories that can be dynamically retrieved under different recommendation contexts.

Finally, together with the aggregated working memory representation, the complete user memory is formulated as:

$$\mathbf{M} = [\mathbf{M}_{long}; \mathbf{s}] \quad (11)$$

Prototype Diversity Regularization To prevent semantic prototypes from collapsing into similar representations, we introduce a diversity regularization term:

$$\mathcal{L}_{ortho} = \frac{1}{K} \sum_{k=1}^K \max_{j \neq k} \cos(\mathbf{p}_k, \mathbf{p}_j), \quad (12)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity. This constraint encourages different memory slots to capture complementary interest patterns.

Decoupled Memory Retrieval

After constructing the hierarchical memory structure, we obtain a user memory representation consisting of working memory and long-term semantic memory. To exploit these memories under different recommendation contexts, the model needs to dynamically retrieve relevant long-term interests according to the current behavioral trajectory.

However, existing sequential recommendation methods typically employ a unified representation space for both memory retrieval and final prediction, where similarity matching and semantic discrimination are optimized simultaneously. Such a design may introduce representation conflicts. To address this issue, we propose Decoupled Memory Retrieval (DMR), which separates memory retrieval and semantic representation learning into two independent spaces. Specifically, DMR first constructs a context-aware query from the complete behavioral trajectory to retrieve relevant semantic memories, and then projects the retrieved information into a dedicated representation space for recommendation prediction.

Context-aware Query Encoding To capture the current recommendation context, we construct a query from the complete behavioral trajectory rather than only recent interactions. However, semantic item representations may lack sufficient discrimination when different items share similar meanings. Therefore, we introduce an independent Trace ID embedding table in the retrieval branch: $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$. For item i , its Trace ID representation is obtained as: $\mathbf{e}_i = \mathbf{E}[i]$.

We inject the Trace ID information into the denoised behavior representation through a gated residual mechanism:

$$g_i = \sigma(\mathbf{W}_g[\mathbf{h}_i; \mathbf{e}_i] + b_g), \quad (13)$$

$$\hat{\mathbf{h}}_i = \text{LN}(\mathbf{h}_i + g_i \odot \mathbf{e}_i), \quad (14)$$

where $\mathbf{h}_i$ denotes the behavior representation produced by WEBD. The trajectory-enhanced sequence is formulated as: $\hat{\mathbf{H}} = [\hat{\mathbf{h}}_1, \hat{\mathbf{h}}_2, \dots, \hat{\mathbf{h}}_L]$. After adding positional embeddings, a causal Transformer is applied:

$$\mathbf{H}_{trace} = \text{CausalTransformer}(\hat{\mathbf{H}} + \mathbf{P}_{trace}). \quad (15)$$

where $\mathbf{P}_{trace}$ denotes the learnable positional embeddings. The hidden state at the last position is projected to obtain the context-aware query: $\mathbf{q}_{trace} = \mathbf{W}_q \mathbf{H}_{trace}[L]$.

This query integrates both long-term behavioral patterns and recent intentions, providing a comprehensive context representation for memory retrieval.

Decoupled Retrieval and Representation Given the context-aware query $\mathbf{q}_{trace}$, DMR retrieves relevant memories from the unified memory bank $\mathbf{M}$. We argue that memory matching and semantic representation learning optimize different objectives. Therefore, DMR explicitly separates these two processes into a Retrieval Space $\mathcal{A}$ and a Representation Space $\mathcal{R}$.

The retrieval space is responsible for measuring the relevance between the current context and historical memories. The memory bank are first projected into the retrieval space:

$$\mathbf{M}^{\mathcal{A}} = \mathbf{W}_m^{\mathcal{A}} \mathbf{M}, \quad (16)$$

where $\mathbf{W}_m^A$ are learnable projection matrices. The retrieval weights are then computed according to the similarity between the context query and memory slots:

$$\beta = \text{softmax} \left(\frac{\mathbf{M}^A(\mathbf{q}^A)^\top}{\sqrt{d_a}} \right), \quad (17)$$

The retrieval space only determines which memories should be selected and does not directly participate in user representation learning. To preserve semantic information for recommendation prediction, the memory bank is independently projected into the representation space:

$$\mathbf{M}^R = \mathbf{W}_m^R \mathbf{M}, \quad (18)$$

where $\mathbf{W}_m^R$ denotes the representation projection matrix. The retrieved memory representation is obtained by aggregating memory slots according to the retrieval weights:

$$\mathbf{r}_{mem} = \beta \mathbf{M}^R. \quad (19)$$

Finally, we combine the retrieved long-term interests with the current behavioral context through gated residual learning:

$$\mathbf{g}_{mem} = \sigma(\mathbf{W}_g[\mathbf{q}_{trace}; \mathbf{r}_{mem}] + b_g), \quad (20)$$

$$\mathbf{z}_u = \text{LN}(\mathbf{q}_{trace} + \mathbf{g}_{mem} \odot \mathbf{r}_{mem}). \quad (21)$$

By decoupling retrieval and representation spaces, DMR allows memory matching and semantic prediction to be optimized independently. This design avoids the interference caused by conflicting objectives in unified representations and produces a more expressive user representation with adaptive coordination between short-term intentions and long-term preferences.

Training Objective

DREAMRec is trained in an end-to-end manner by jointly optimizing the recommendation objective and auxiliary regularization objectives. Specifically, we consider next-item prediction loss, contrastive learning loss, semantic prototype diversity regularization, and frequency consistency regularization.

Recommendation Loss Given the learned user representation $\mathbf{z}_u$, we adopt sampled softmax cross-entropy for next-item prediction. Given the positive item v^+ and sampled negative items $\mathcal{N}$, the recommendation loss is defined as:

$$\mathcal{L}_{rec} = -\log \frac{\exp(s(u, i^+)/\tau)}{\exp(s(u, i^+)/\tau) + \sum_{i^- \in \mathcal{N}} \exp(s(u, i^-)/\tau)}, \quad (22)$$

where $s(u, v) = \mathbf{z}_u^T \mathbf{z}_v$ denotes the matching score between user and item representations, and τ is the temperature coefficient.

Contrastive Learning Objective To improve the robustness of user representations against behavioral perturbations, we introduce a contrastive learning objective. For each user sequence, a perturbed view is generated by randomly masking historical interactions, and the corresponding user representation $\tilde{\mathbf{z}}_u$ is obtained using the same encoder.

We adopt a symmetric InfoNCE objective, treating different views of the same user as positive pairs and other users within the mini-batch as negative samples:

$$\mathcal{L}_{CL} = -\frac{1}{2}(\mathcal{L}_{u \rightarrow \tilde{u}} + \mathcal{L}_{\tilde{u} \rightarrow u}), \quad (23)$$

where $\mathcal{L}_{u \rightarrow \tilde{u}} = -\log \frac{\exp(\langle \tilde{\mathbf{z}}_u, \hat{\mathbf{z}}_u \rangle / \tau)}{\sum_{j \in \mathcal{B}} \exp(\langle \tilde{\mathbf{z}}_u, \hat{\mathbf{z}}_j \rangle / \tau)}$, and $\mathcal{L}_{\tilde{u} \rightarrow u}$ is defined symmetrically. Here, $\hat{\mathbf{z}} = \mathbf{z} / \|\mathbf{z}\|_2$ denotes the normalized representation and $\mathcal{B}$ represents the mini-batch user set.

Overall Objective The final optimization objective is formulated as:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_{CL} \mathcal{L}_{CL} + \lambda_{proto} \mathcal{L}_{proto} \quad (24)$$

where λ_{CL} and λ_{proto} balance the contributions of the auxiliary objectives.

Experiments

Experimental Settings

Datasets We evaluate DREAMRec on four public sequential recommendation datasets, including three Amazon Review 2023 datasets (Amazon-Baby, Amazon-Instrument, and Amazon-Game) and MovieLens-1M (ML-1M). These datasets cover diverse recommendation scenarios with different sparsity levels and sequence characteristics.

For Amazon datasets, we concatenate item title, brand, category, attributes, and description as textual input and obtain semantic representations using a pre-trained T5 encoder. For ML-1M, movie titles and categories are used as textual information. Following previous works [10, 36, 42, 43], we apply the 5-core filtering strategy to Amazon datasets. All user interactions are chronologically ordered and split using the leave-one-out protocol [23, 42], where the last interaction is used for testing, the second last interaction for validation, and the remaining interactions for training.

Baselines We compare DREAMRec with representative sequential recommendation baselines, including traditional methods GRU4Rec [8], SASRec [14], and BERT4Rec [27]; semantic-enhanced methods FDSA [39], S3Rec [42], UniSRec [11], VQRec [9], MMSR [12], and CCFRec [18]; robust sequential recommendation method FMLP-Rec [43]; and self-supervised methods DuoRec [21] and MAERec [38].

Evaluation Metrics Following previous studies [11, 18, 36, 42], we adopt the full-ranking evaluation protocol and report Recall@K and NDCG@K with $K \in \{5, 10\}$.

Implementation Details DREAMRec is implemented in PyTorch and optimized using AdamW. The embedding dimension and hidden dimension are set to 128 and 512, respectively. The learning rate is tuned from $\{5 \times 10^{-4}, 1 \times 10^{-3}\}$, while other baselines follow their original settings.

For DREAMRec, we jointly optimize the recommendation loss, contrastive learning objective, and regularization terms. The batch size is set to 300, with $\tau = 0.05$, $\lambda_{CL} = 0.2$, and $\lambda_{proto} = 0.1$. All hyperparameters are selected on the validation set, and early stopping with patience 10 is adopted.

Methods	Baby				Game				Scientific				ML-1M			
	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10
BERT4Rec	1.76	3.05	1.21	1.49	4.65	7.30	3.03	3.81	1.91	2.91	1.24	1.50	13.93	24.18	9.64	12.64
GRU4Rec	2.25	3.49	1.45	1.79	5.35	8.15	3.55	4.38	2.35	3.69	1.53	1.89	13.16	21.54	7.25	9.52
SASRec	2.16	3.57	1.40	1.75	5.40	8.52	3.36	4.33	2.64	4.07	1.55	1.94	14.05	23.00	8.74	11.46
FMLPRec	2.33	3.62	1.51	1.85	5.23	8.62	3.43	4.39	2.64	4.17	1.60	1.99	14.14	23.63	9.41	12.34
FDSA	2.38	3.92	1.63	2.00	5.49	8.57	3.56	4.65	2.68	4.21	1.78	2.24	13.97	22.86	8.92	11.70
S ³ Rec	2.21	3.33	1.31	1.73	4.90	7.74	3.20	4.11	2.58	4.23	1.76	2.14	13.86	22.68	8.68	11.38
DuoRec	2.06	3.51	1.44	1.77	5.64	8.49	3.73	4.64	2.50	3.84	1.61	2.14	14.17	22.81	9.68	11.17
MAERec	2.19	3.75	1.41	1.88	6.23	9.31	4.06	5.08	2.98	4.57	2.10	2.48	13.97	23.45	9.42	11.65
UniSRec	2.34	3.81	1.45	1.85	5.68	9.16	3.52	4.64	2.81	4.62	1.62	2.09	13.79	22.57	8.69	11.40
VQRec	2.65	4.04	1.71	2.09	5.86	9.21	3.60	4.71	2.98	4.56	1.75	2.19	13.29	24.06	9.82	12.80
MMSR	2.27	3.94	1.37	1.90	5.63	8.76	3.77	4.56	2.69	4.22	1.75	2.03	13.38	24.11	9.59	12.09
CCFRec	<u>2.81</u>	<u>4.60</u>	<u>1.79</u>	<u>2.43</u>	<u>6.63</u>	<u>10.37</u>	<u>4.08</u>	<u>5.41</u>	<u>3.69</u>	<u>5.50</u>	<u>2.29</u>	<u>2.80</u>	<u>14.24</u>	<u>24.37</u>	<u>9.09</u>	<u>12.13</u>
Ours	3.09	4.92	1.97	2.55	7.01	11.37	4.22	5.58	3.96	6.04	2.44	3.12	16.65	26.11	10.42	13.44
Improv.	+10.0%	+7.0%	+10.1%	+4.9%	+5.7%	+9.6%	+3.4%	+3.1%	+7.3%	+9.8%	+6.6%	+11.4%	+16.9%	+7.1%	+14.6%	+10.8%

Table 1: Performance comparisons among different methods (%). The columns are ordered as Baby, Game, Scientific, and ML-1M. “Ours” denotes the results of DREAMRec. Best results are highlighted in **bold** and second-best results are underlined.

Variants	Game				Scientific				ML-1M			
	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10
(0) DREAMRec	7.01	11.37	4.22	5.58	3.96	6.04	2.44	3.12	16.65	26.11	10.42	13.44
(1) w/o wavelet	6.95	11.22	4.13	5.49	3.89	5.98	2.31	3.04	15.69	25.94	10.04	13.01
(2) fixed threshold	6.88	10.99	4.06	5.41	3.72	5.83	2.04	2.71	15.77	25.59	9.85	12.89
(3) w/o memory	6.91	11.18	4.11	5.48	3.80	5.96	2.30	2.99	15.84	25.34	9.54	12.87
(4) w/o SMP	6.88	11.05	4.01	5.38	3.55	5.32	2.06	2.51	15.66	25.48	9.86	12.92
(5) w/o decoupling	6.86	11.07	3.89	5.32	3.86	5.73	2.27	2.98	15.57	25.33	9.49	12.62

Table 2: Ablation study of our proposed method on three datasets. The best and second-best results are denoted in bold and underlined fonts, respectively.

Overall Performance

As shown in Table 1, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

The performance gains mainly stem from the progressive modeling strategy of DREAMRec. Specifically, WEBD suppresses random behavioral noise while preserving informative bursty interests, Semantic Memory Modeling organizes redundant historical interactions into compact semantic memories, and DMR decouples memory retrieval from representation learning to alleviate the optimization conflict caused by unified user representations. These components work collaboratively to produce more robust and expressive user representations.

Further analysis on different datasets reveals that DREAMRec consistently outperforms existing methods on

all three Amazon datasets, demonstrating its ability to alleviate the representation limitations caused by sparse user-item interactions. More importantly, DREAMRec achieves substantially larger improvements on the long-sequence ML-1M dataset, suggesting that the proposed semantic memory and context-aware retrieval mechanisms are particularly effective in modeling long-term interest evolution and retrieving relevant historical preferences under complex recommendation scenarios.

Ablation Study

As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

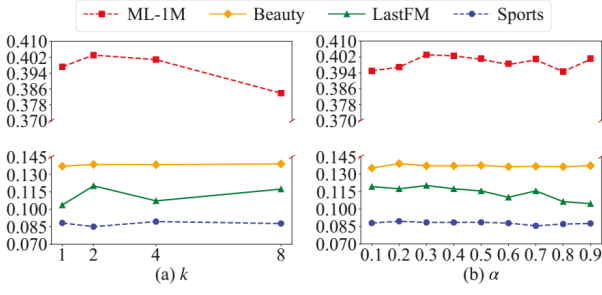


Figure 3: Visualization of the learned adaptive G and spectral changes in the FDFR module.

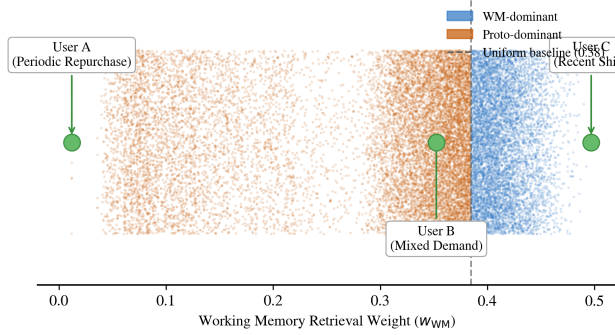


Figure 4: Visualization of the learned adaptive G and spectral changes in the FDFR module.

Hyper-parameter analysis

wm length As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

n prototypes As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

Further Analysis

case1-final As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn

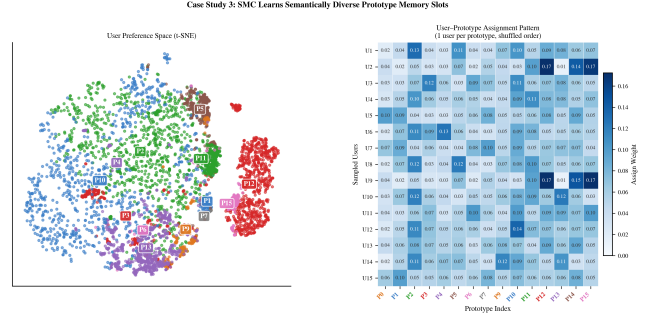


Figure 5: Visualization of the learned adaptive G and spectral changes in the FDFR module.

more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

case3-final As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

Related work

case1-final As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

case3-final As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

conclusion

As shown in Table 2, DREAMRec consistently achieves the best performance across all datasets and evaluation metrics, demonstrating the effectiveness of the proposed user modeling framework. Compared with the strongest competing baseline, CCFRec, DREAMRec consistently improves both Recall and NDCG, indicating its ability to learn more accurate and discriminative user representations while generalizing well across datasets with different characteristics.

References